TABLE III: Effect of the modification of the Perch front-end configuration on classification performance measured on the ATBFL dataset, using the multiclass configuration with $k = 5$. The frame length, the frequency range spanned by the mel filterbank, and the resulting macro-averaged AUC-ROC and AP are shown.

Frame length	Frequency (Hz) Min	Max	AUC-ROC	AP
640†	60	16000	0.636	0.146
640	0	16000	0.647	0.146
1024	0	16000	0.620	0.116
2048	0	16000	0.534	0.035
4096	0	16000	0.538	0.035
8192	0	16000	0.558	0.064
16384	0	16000	0.564	0.064
32768	0	16000	0.569	0.065
640	0	1000	0.618	0.082
1024	0	1000	0.629	0.088
2048	0	1000	0.664	0.106
4096	0	1000	0.683	0.149
8192	0	1000	**0.690**	**0.162**
16384	0	1000	0.682	0.161
32768	0	1000	0.688	0.150

† Default input representation of Perch.

A. Perch input representation

Table III presents the effect of the adjustments made to the Perch front-end, evaluated on the ATBFL dataset in the multiclass configuration with $k = 5$. Lowering the lower edge of the mel filterbank from 60 Hz to 0 Hz, while retaining the default frame length, improves the AUC-ROC by $1.1\,\%$ absolute and leaves the AP unchanged. Although the default configuration discards the fundamental frequency of most call types, the recovered energy is still spread over too few mel bands to be of substantial benefit to the model.

Once the filterbank is restricted to lie between 0 Hz and 1000 Hz, performance improves consistently with increasing frame length, peaking at 8192 samples (256 ms at 32 kHz) with an AUC-ROC of 0.690 and an AP of 0.162. This corresponds to absolute improvements of $5.4\,\%$ and $1.6\,\%$ over the default configuration. No further gains are observed beyond 8192 samples. This configuration is used for all subsequent experiments, and is referred to as *Perch + adjustment*. It was selected on the ATBFL dataset and is applied without further modification to the FB dataset.

B. ATBFL dataset

Table IV presents the per-class and macro-averaged results obtained on the ATBFL dataset for $k = 1$, 5 and 10. The adjusted input representation improves the macro AUC-ROC at every support set size, with absolute gains of $4.0\,\%$, $5.4\,\%$ and $5.8\,\%$ respectively. Almost all of this gain arises from the `bp` call, whose AUC-ROC rises from 0.783 to 0.950 at $k = 10$, an absolute improvement of $16.7\,\%$. The AUC-ROC of the `d` call also improves, by $3.9\,\%$ absolute, while that of the `bmabz` call falls by $3.4\,\%$ absolute.

The multilabel configuration recovers the `bmabz` call to an AUC-ROC of 0.607, above the 0.579 obtained with the unmodified system, and achieves the highest macro AUC-ROC at every value of k, reaching 0.739 at $k = 10$. The improvement over the multiclass configuration is concentrated in the `bmabz` and `d` calls, rising by $6.2\,\%$ and $4.4\,\%$ absolute respectively. The multiclass configuration nevertheless retains a small improvement in AP, achieving 0.176 against 0.174 at $k = 10$. The multilabel deficit is confined entirely to the `bp` call, whose AP falls from 0.350 to 0.295, while both `bmabz` and `d` improve in AP as well as in AUC-ROC. Taken together, the adjusted input representation and the multilabel formulation yield a $9.1\,\%$ absolute improvement in macro AUC-ROC over the unmodified Perch system at $k = 10$, a relative improvement of $14.0\,\%$.

Figure 2 shows how performance varies as the support set size is increased from $k = 1$ to $k = 10$. All three configurations improve with k in terms of both metrics over the range considered. Of the $9.8\,\%$ total absolute gain in AUC-ROC achieved by the multilabel configuration, $7.9\,\%$ is obtained between $k = 1$ and $k = 5$. The multilabel configuration attains the highest macro AUC-ROC at every value of k considered, and the adjusted multiclass configuration the highest AP, although the multilabel AP falls below that of the unmodified Perch system at $k = 1$.

For the ATBFL data, we see that none of the few-shot systems approach the fully supervised WhaleVAD topline, which achieves an AUC-ROC of 0.808 and an AP of 0.36. At $k = 10$, a difference of $7\,\%$ in AUC-ROC and $19\,\%$ in AP remains. The absolute AP values are low throughout, but must be interpreted with the class imbalance of the dataset in mind. The expected AP of an uninformative ranking is equal to the positive rate of the class, whereas the expected AUC-ROC is always 0.5. Annotated calls occupy only $3.49\,\%$ of the ATBFL recordings, and the AP of 0.174 achieved by the best few-shot system therefore remains well above chance, as does the 0.36 achieved by the topline. The two metrics consequently convey different information, and AP in particular cannot be compared across classes or datasets whose positive rates differ.

Figure 3 shows the per-class ROC curves obtained for the two ATBFL subsets, comparing the best few-shot system at $k = 10$ with WhaleVAD. WhaleVAD retains a clear advantage for the `bmabz` and `d` calls on both subsets. For the `bp` call, however, the few-shot classifier is superior.

While the fully supervised WhaleVAD topline remains ahead, the margin is modest given that the prototypical classifier observes at most ten labelled examples per call type, while WhaleVAD was trained on the full ATBFL training set containing at least 1646 examples per call type and over 58 000 examples in total.

C. FB dataset

Figure 4a shows the performance achieved by the prototypical classifier in conjunction with the two embedding networks when applied to the FB dataset and when the task is reduced to binary detection, where a segment is considered positive if it contains any Bryde's whale vocalisation. Perch markedly outperforms WhaleVAD across all values of k, achieving an